Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9738219895287958	0.968421052631579 – 0.9738219895287958

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.18717663909268947	0.1865605658709107 – 0.18858212447356107

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.22232916265640038	0.22212072115550346 – 0.22289301685586516

Thin QR

512 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.7779124044095739	0.7753794875913581 – 0.7799979937807202

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.578889476761995	0.5775322067599985 – 0.5826379763927365

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.8276896699528504	0.8116910229645093 – 0.8392311052861512

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.8266666666666667	0.8089117395029991 – 0.8577699736611062

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9572822465492623	0.9372616558002215 – 1.021105141980046

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1	1 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9989429175475687	0.6957478005865103 – 1.4261603375527425

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.6532593619972261 – 1.0021276595744681

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	**0.9964324917672887**	95% INTERVAL	0.8442432683379758 – 1.0080343537886134

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9974604966139955	0.843441466854725 – 1.0025359256128488

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1	0.9791666666666666 – 1.0208333333333333

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9978308026030369	0.9956772334293948 – 1.032490974729242

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9919571045576407	0.908344733242134 – 1.0027359781121752

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	1 – 1.28

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	1 – 1

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0008684324793746	0.9608865710560626 – 1.0411206507004067

Cholesky factorization

256 × 256 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0045150014564521	1.0007302468234263 – 1.0139275766016713

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0013484710078733	0.979931106784484 – 1.0090029357399153

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0096153846153846	0.9894490035169988 – 1.0169491525423728

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0204508482454102	0.7212323064113239 – 1.3266775017413512

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0017421602787457	0.9702797202797203 – 1.2707993474714518